Grade 10 Term 3 — Learning Evidence 1 Exam Solutions (C5)

Evaluated criterion

C5. Uses the standard normal table and graphical interpretation to determine probabilities for normal distributions.

Standard normal table (values of $\Phi(z)$). Use this table for all questions below.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

1 Problem 1 — Left area and right tail

Problem. Let $Z \sim N(0,1)$.

Tasks. Calculate $P(Z < 0.48)$ and $P(Z > 1.25)$.

Solution

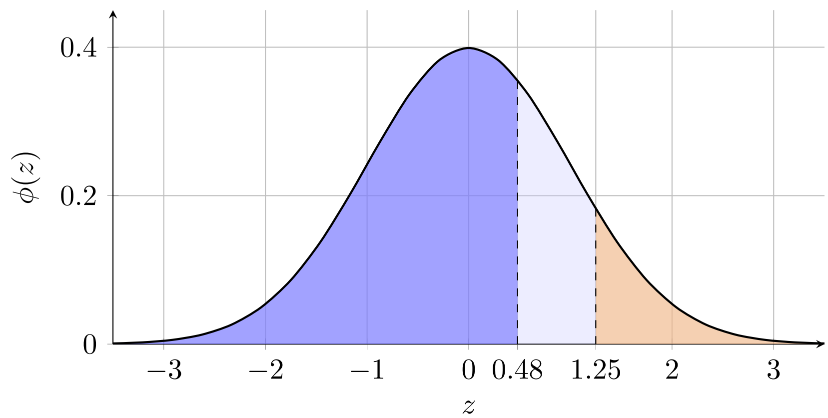
From the table,

$$\Phi(0.48) = 0.6844, \qquad \Phi(1.25) = 0.8944.$$

So

$$P(Z < 0.48) = 0.6844,$$

$$P(Z > 1.25) = 1 - \Phi(1.25) = 1 - 0.8944 = 0.1056.$$



Interpretation. About 68.4% of values are less than 0.48, and about 10.6% are greater than 1.25.

2 Problem 2 — Interval crossing zero

Problem. Let $Z \sim N(0, 1)$.

Task. Find $P(-0.85 \leq Z \leq 1.30)$.

Solution

From the table,

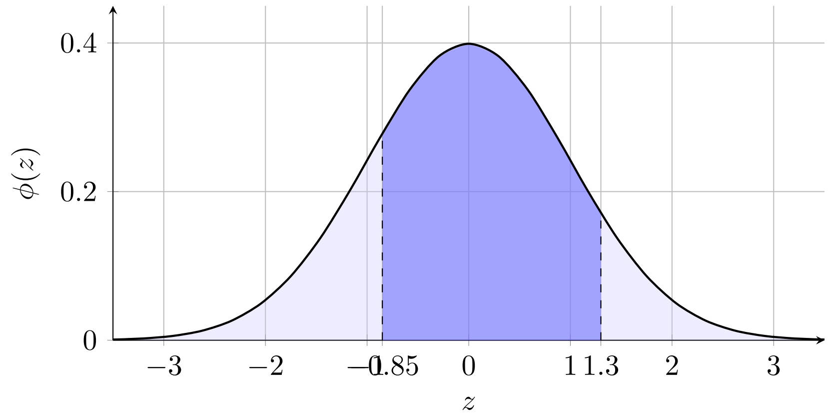
$$\Phi(1.30) = 0.9032, \qquad \Phi(0.85) = 0.8023.$$

By symmetry,

$$\Phi(-0.85) = 1 - \Phi(0.85) = 1 - 0.8023 = 0.1977.$$

Therefore,

$$P(-0.85 \leq Z \leq 1.30) = \Phi(1.30) - \Phi(-0.85) = 0.9032 - 0.1977 = 0.7055.$$



Interpretation. About 70.6% of values are between −0.85 and 1.30.

3 Problem 3 — Two-tail union

Problem. Let $Z \sim N(0, 1)$.

Task. Determine $P(Z \leq -1.35 \text{ or } Z \geq 0.75)$.

Solution

From the table,

$$\Phi(1.35) = 0.9115, \qquad \Phi(0.75) = 0.7734.$$

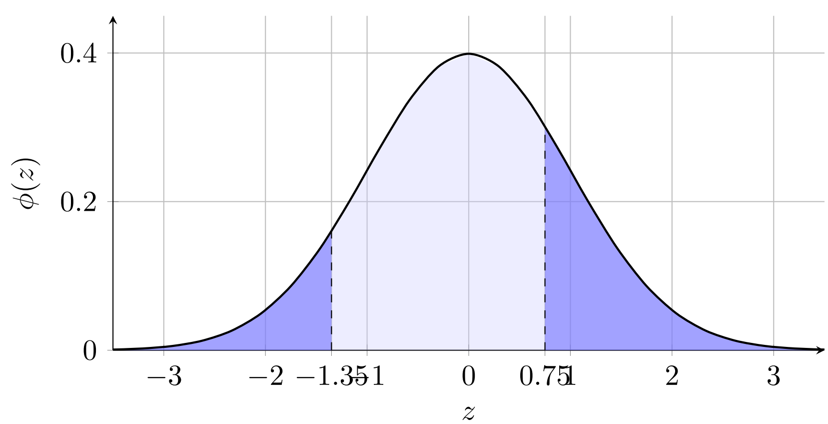
Thus,

$$P(Z \leq -1.35) = 1 - \Phi(1.35) = 1 - 0.9115 = 0.0885,$$

$$P(Z \geq 0.75) = 1 - \Phi(0.75) = 1 - 0.7734 = 0.2266.$$

The events are disjoint, so

$$P(Z \leq -1.35 \text{ or } Z \geq 0.75) = 0.0885 + 0.2266 = 0.3151.$$



Interpretation. About 31.5% of values fall in the two outer regions.

4 Problem 4 — Absolute value and complement

Problem. Let $Z \sim N(0,1)$.

Tasks. Compute $P(|Z| \le 1.50)$ and $P(|Z| > 1.50)$.

Solution

From the table,

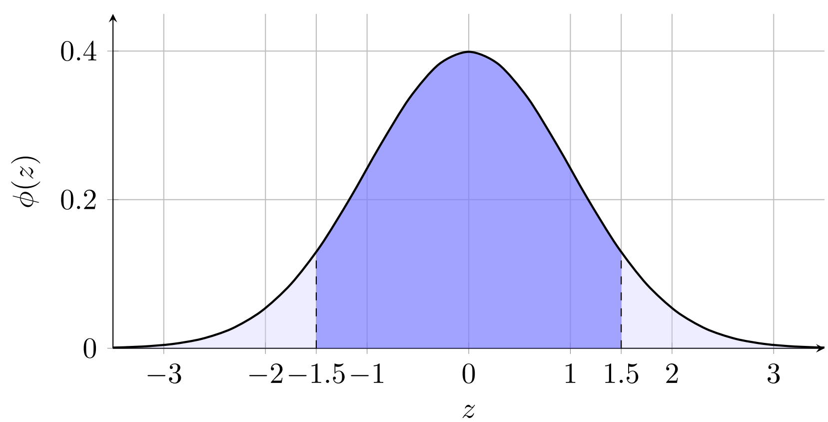
$$\Phi(1.50) = 0.9332, \qquad \Phi(-1.50) = 1 - 0.9332 = 0.0668.$$

Then

$$P(|Z| \le 1.50) = P(-1.50 \le Z \le 1.50) = 0.9332 - 0.0668 = 0.8664.$$

So

$$P(|Z| > 1.50) = 1 - 0.8664 = 0.1336.$$



Interpretation. Around 86.6% of values are within 1.50 standard units of 0, and 13.4% are outside.

5 Problem 5 — Between two positive bounds and upper tail

Problem. Let $Z \sim N(0,1)$.

Tasks. Find $P(0.60 < Z < 1.80)$ and $P(Z \ge 1.80)$.

Solution

From the table,

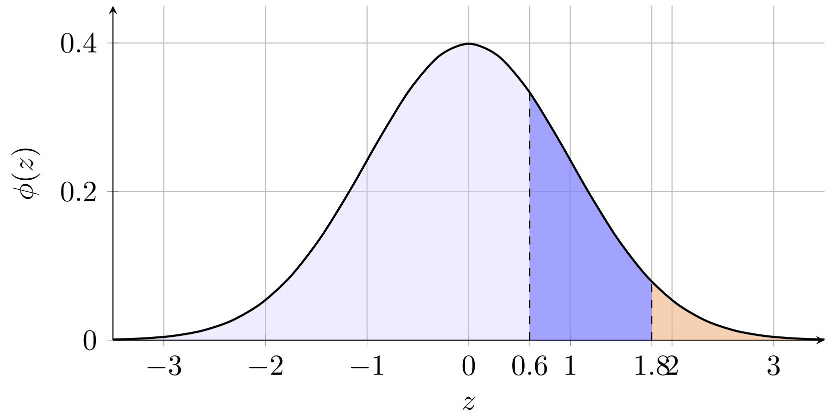
$$\Phi(1.80) = 0.9641, \qquad \Phi(0.60) = 0.7257.$$

Hence

$$P(0.60 < Z < 1.80) = \Phi(1.80) - \Phi(0.60) = 0.9641 - 0.7257 = 0.2384.$$

And

$$P(Z \ge 1.80) = 1 - \Phi(1.80) = 1 - 0.9641 = 0.0359.$$



Interpretation. About 23.8% of values lie in that band, and only 3.6% lie above 1.80.

6 Problem 6 — Symmetry and decomposition

Problem. Let $Z \sim N(0, 1)$.

Tasks. Find $P(Z < -1.10)$ and $P(-1.10 \leq Z \leq 0)$.

Solution

From the table,

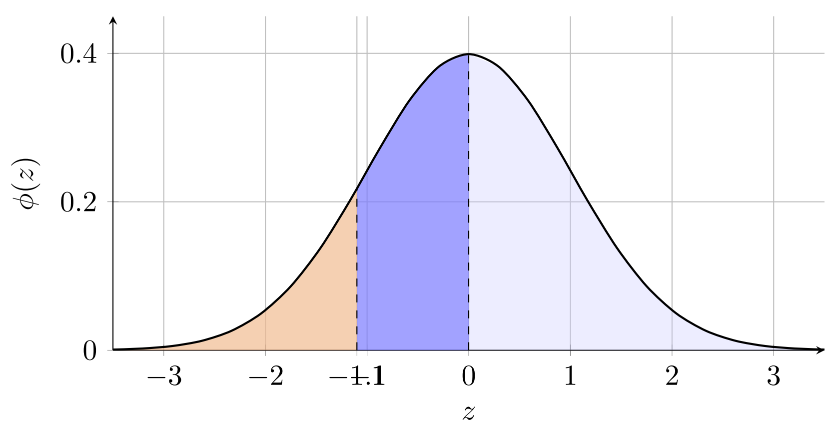
$$\Phi(1.10) = 0.8643, \qquad \Phi(0) = 0.5000.$$

By symmetry,

$$P(Z < -1.10) = \Phi(-1.10) = 1 - \Phi(1.10) = 1 - 0.8643 = 0.1357.$$

Then

$$P(-1.10 \leq Z \leq 0) = \Phi(0) - \Phi(-1.10) = 0.5000 - 0.1357 = 0.3643.$$



Interpretation. About 13.6% of values are below −1.10, and 36.4% lie between −1.10 and 0.

Conclusion. In every question, the same method is used: identify the required region under the normal curve, translate it into Φ-values from the table, apply symmetry and complements when needed, and interpret the probability in context.